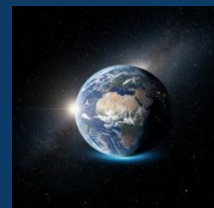




QUALITY IN SPORT

eISSN 2450-3118 · Open Access · Peer-reviewed

apcz.umk.pl/QS Nicolaus Copernicus University in Toruń



IWASZKO, Magdalena Maria, SKAJEWSKA, Marcelina, ZIĘBKA, Wojciech, BOJANOWSKI, Szymon and WOJTERA, Aleksandra. The Role of Vitamin D Deficiency in the Pathogenesis and Course of Selected Autoimmune and Immune - Mediated Diseases. *Quality in Sport*. 2026;73:75365. <https://doi.org/10.12775/QS.2026.73.75365>

ARTICLE TIMELINE

Received: 16.08.2026. Accepted: 12.09.2026. Published: 17.09.2026.

The journal has been awarded 20 points in the parametric evaluation by the Polish Ministry of Higher Education and Science (Annex to the announcement of 05.01.2024, No. 32553). Unique Journal Identifier: 201398. Scientific disciplines assigned: Economics and Finance (Field of Social Sciences); Management and Quality Sciences (Field of Social Sciences).

Punkty Ministerialne z 2019 – aktualny rok 20 punktów. Załącznik do komunikatu Ministra Szkolnictwa Wyższego i Nauki z dnia 05.01.2024 Lp. 32553. Posiada Unikatowy Identyfikator Czasopisma: 201398. Przypisane dyscypliny naukowe: Ekonomia i finanse (Dziedzina nauk społecznych); Nauki o zarządzaniu i jakości (Dziedzina nauk społecznych). © The Authors 2026.

OPEN ACCESS · CC BY-NC-SA 4.0 This article is published with open access under the License Open Journal Systems of Nicolaus Copernicus University in Toruń, Poland, and is distributed under the terms of the Creative Commons Attribution Non-commercial Share Alike License (<http://creativecommons.org/licenses/by-nc-sa/4.0/>), which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the work is properly cited. The authors declare no conflict of interest regarding the publication of this paper.

The Role of Vitamin D Deficiency in the Pathogenesis and Course of Selected Autoimmune and Immune - Mediated Diseases

Magdalena Iwaszko

Jan Bziel University Hospital No. 2 in Bydgoszcz, Poland

ORCID: <https://orcid.org/0009-0000-6010-5443>

E-mail: iwaszkomagda1102@gmail.com

Marcelina Skajewska

Jan Bziel University Hospital No. 2 in Bydgoszcz, Poland

ORCID: <https://orcid.org/0009-0009-5029-192X>

E-mail: msmarcelina49@gmail.com

Wojciech Ziębka

A. Falkiewicz Specialist Hospital in Wrocław, Poland

ORCID: <https://orcid.org/0009-0000-6835-6773>

E-mail: ziebka.wojciech@gmail.com

Szymon Bojanowski

Medical University of Lublin, Poland

ORCID: <https://orcid.org/0009-0005-4264-102X>

E-mail: szymbojanv2@gmail.com

Aleksandra Wojtera

Raszeja City Hospital in Poznań, Poland

ORCID: <https://orcid.org/0009-0007-7288-676X>

E-mail: owojtera74@gmail.com

Corresponding Author : Magdalena Iwaszko

E-mail : iwaszkomagda1102@gmail.com

Abstract

Introduction: Vitamin D has immunomodulatory effects, which may be relevant in autoimmune and immune-mediated diseases.

Aim: The aim of this study was to present current data on the role of vitamin D deficiency in the pathogenesis and course of selected diseases.

Methods: A narrative literature review was conducted, including mainly publications from 2018–2026 concerning multiple sclerosis, rheumatoid arthritis, autoimmune thyroid diseases, and inflammatory bowel diseases.

Results: Vitamin D deficiency is frequently observed in the analyzed diseases. Mechanistic data indicate its influence on dendritic cells, T and B lymphocytes, and the regulation of the inflammatory response. However, the results of supplementation studies remain heterogeneous; some studies have shown improvement in selected laboratory or radiological parameters or disease activity indices.

Conclusions: Available data do not allow vitamin D deficiency to be unequivocally considered a causal factor, nor supplementation to be regarded as a specific immunomodulatory treatment. Correction of confirmed deficiency remains justified, while the effect of supplementation on the long-term course of these diseases requires further research.

Keywords: vitamin D, vitamin D deficiency, 25-hydroxyvitamin D, autoimmunity, multiple sclerosis, rheumatoid arthritis, autoimmune thyroid disease, inflammatory bowel disease, Crohn's disease, ulcerative colitis

1. Introduction

Autoimmune and immune-mediated diseases constitute a heterogeneous group whose common feature is dysfunction of the immune system. Autoimmune diseases are characterized by a loss of immunological tolerance to self-antigens; this phenomenon results in the activation of autoreactive immune cells and the development of a response directed against the body's own tissues and organs. Genetic predisposition, environmental factors, and abnormalities in the mechanisms responsible for regulating the immune response are the main interacting components of the complex process leading to the development of autoimmunity [1,2].

Vitamin D is one of the environmental factors that may influence the regulation of the autoimmune response. The effects of vitamin D extend beyond the regulation of calcium-phosphate homeostasis and bone tissue metabolism [1,2,3]. The active form of vitamin D -

1,25-dihydroxyvitamin D [$1,25(\text{OH})_2\text{D}$] - has immunomodulatory properties, affecting both innate and adaptive immunity. The vitamin D receptor (VDR) is present in numerous immune cells, including monocytes, macrophages, dendritic cells, and T and B lymphocytes [1,2,3,4].

Vitamin D deficiency is common in the general population; its prevalence depends, among other factors, on latitude, season, and age [5,6]. In patients with chronic inflammatory and autoimmune diseases [7], including multiple sclerosis [4,8], rheumatoid arthritis [9], systemic lupus erythematosus [10], and inflammatory bowel disease [11], deficiency of 25-hydroxyvitamin D [$25(\text{OH})\text{D}$] is common.

The interpretation of these associations should, however, remain cautious. Low $25(\text{OH})\text{D}$ concentrations and their correlation with disease activity do not constitute evidence of a causal relationship [7,9]. Vitamin D deficiency may be involved in the pathogenesis of immune disorders, but it also cannot be excluded that it is a consequence of chronic inflammation, reduced physical activity, or lower exposure to sunlight. The final assessment of the significance of vitamin D should be based on both biological mechanisms and the results of epidemiological and clinical studies [12,13].

The aim of this paper is to present the current state of knowledge on the role of vitamin D deficiency in the pathogenesis and course of selected autoimmune and immune-mediated diseases, with particular emphasis on multiple sclerosis, rheumatoid arthritis, autoimmune thyroid diseases, and inflammatory bowel disease.

2. Literature review methodology

This paper is a narrative review of the literature concerning the role of vitamin D deficiency in the pathogenesis and course of selected autoimmune and immune-mediated diseases. To collect the material, scientific publications were analyzed using the PubMed electronic database. Keywords and their combinations were used, including: vitamin D, vitamin D deficiency, 25-hydroxyvitamin D, autoimmunity, multiple sclerosis, rheumatoid arthritis, autoimmune thyroid disease, inflammatory bowel disease, Crohn's disease, and ulcerative colitis. Publications from 2018-2026 were analyzed; particular attention was paid to the most recent data concerning the relationship between $25(\text{OH})\text{D}$ concentrations and the risk of occurrence, activity, and course of the analyzed diseases. Systematic reviews and meta-analyses, observational and prospective studies, randomized clinical trials, as well as current guidelines and expert statements were included in the analysis. Particular importance was given to studies that allowed assessment of both the association between vitamin D deficiency and disease course and the potential effects of its supplementation.

3.1. Vitamin D metabolism

Cutaneous synthesis and diet together with supplements are the two routes by which vitamin D is supplied to the body. UVB radiation causes the conversion of 7-dehydrocholesterol in the skin to previtamin D_3 , which then undergoes isomerization to cholecalciferol (vitamin D_3). Vitamin D may also be supplied in the form of cholecalciferol or ergocalciferol (vitamin D_2) [3,4]. It then undergoes a two-step hydroxylation - initially in the liver to 25-hydroxyvitamin D [$25(\text{OH})\text{D}$], and subsequently, mainly in the kidneys, to biologically active 1,25-dihydroxyvitamin D [$1,25(\text{OH})_2\text{D}$], i.e. calcitriol [3].

Serum $25(\text{OH})\text{D}$ concentration is the primary parameter used to assess vitamin D status in the body [3]. The active form of vitamin D acts primarily through the vitamin D receptor (VDR), which belongs to the nuclear receptor family. VDR is present not only in tissues associated with calcium-phosphate homeostasis, but also in immune cells, including monocytes, macrophages, dendritic cells, and T and B lymphocytes [1,2,4]. It is noteworthy that some

immune cells are capable of expressing the CYP27B1 enzyme; this process enables the local conversion of 25(OH)D to active 1,25(OH)₂D [1,4].

3.2. Effect of vitamin D on innate immunity

Vitamin D affects both types of immunity - innate and adaptive. Within innate immunity, it acts on monocytes and macrophages, modulating their activity and antimicrobial defense mechanisms [1,14]. At the same time, it affects dendritic cells by limiting their maturation and ability to activate T lymphocytes. This promotes the development of a more tolerogenic immune environment [1,2].

In the study by Hafkamp et al., exposure of human dendritic cells to calcitriol [1,25(OH)₂D₃] reduced the production of cytokines promoting the differentiation of Th1 and Th17 lymphocytes, including IL-12 and IL-23. At the same time, increased differentiation of regulatory Treg lymphocytes and IL-10-producing cells was observed. Similar results were obtained by Nagy et al., who demonstrated that dendritic cells exposed to the active metabolite of vitamin D₃ promoted the development of regulatory CD4⁺ lymphocytes while limiting the differentiation of Th1 and Th17 cells [15,16].

3.3. Effect of vitamin D on adaptive immunity and immunological tolerance

An important element of the immunomodulatory action of vitamin D is its effect on the balance between pro-inflammatory and regulatory T-lymphocyte subpopulations. The active form of vitamin D may limit responses associated with Th1 and Th17 lymphocytes and support the development and function of Treg lymphocytes [15,16,17]. This may be relevant in autoimmune diseases because excessive activity of the Th1 and Th17 pathways and insufficient regulatory control are involved in maintaining a chronic inflammatory response.

Vitamin D also affects B lymphocyte function. Available data indicate that it may limit their proliferation, differentiation toward plasma cells, and immunoglobulin production, which may be relevant in diseases associated with autoantibody production [17]. These effects should, however, be interpreted primarily as biological mechanisms that may explain the observed clinical associations, rather than as evidence that vitamin D deficiency itself leads to the development of autoimmune disease.

In summary, vitamin D affects numerous elements of innate and adaptive immunity, promoting a more regulatory and tolerogenic immune response profile. Vitamin D deficiency may therefore theoretically contribute to weakening the mechanisms that control an excessive inflammatory response. However, the significance of these mechanisms for the development and course of specific diseases requires evaluation in light of clinical data, because the observed association between low 25(OH)D concentrations and autoimmunity is not equivalent to a confirmed causal relationship [7,17].

4.1. Multiple sclerosis

Multiple sclerosis (MS) is a chronic inflammatory demyelinating disease of the central nervous system, in the development of which both genetic predisposition and environmental factors play an important role. Vitamin D deficiency is one of the environmental factors being investigated. Low 25(OH)D concentrations have been associated both with the risk of developing MS and with greater disease activity [18,19]. In the study by Giordano et al., vitamin D concentration before initiation of disease-modifying therapy was associated with the subsequent risk of disease activity. The authors also observed that a genetic predisposition to higher vitamin D concentrations was associated with a lower risk of disease activity and a later age at disease onset [18,19]. These findings support the hypothesis of a possible role of vitamin D in the pathogenesis and course of MS; however, by themselves they do not constitute evidence of the efficacy of its supplementation [18].

The results of interventional studies remain inconclusive. The randomized SOLAR trial evaluated the use of high-dose cholecalciferol as adjunctive treatment in patients with relapsing-remitting MS receiving interferon β -1a. After 48 weeks, no significant difference was found in the primary endpoint, which was no evidence of disease activity defined as NEDA-3. However, more favorable results were observed for selected MRI parameters, including a lower number of combined unique active lesions in the group receiving high-dose vitamin D₃ [20]. These findings suggested a possible effect of supplementation on selected radiological parameters despite the lack of a significant effect on the primary endpoint [20].

Similar clinical benefits were not confirmed in the VIDAMS trial. Patients with relapsing-remitting MS treated with glatiramer acetate received either 5000 IU or 600 IU of vitamin D₃ daily for 96 weeks. The proportion of patients who experienced a confirmed relapse did not differ significantly between the high-dose and low-dose vitamin D₃ groups. Thus, the study did not confirm that the use of higher doses of vitamin D as adjunctive therapy reduced the risk of relapse in patients with established relapsing-remitting MS [21].

More recent data were provided by the D-Lay MS study published in 2025, which included patients with a recently diagnosed clinically isolated syndrome typical of MS, some of whom also met the current criteria for relapsing-remitting MS. Cholecalciferol at a dose of 100,000 IU every 2 weeks for up to 24 months reduced the risk of the composite endpoint including relapse or new MRI activity compared with placebo (HR 0.66; 95% CI 0.50-0.87) [19]. Significant differences in favor of cholecalciferol were observed for MRI parameters, whereas the frequency of clinical relapses did not differ significantly between the groups [19].

Current data therefore indicate a biologically and clinically significant association between vitamin D status and MS, although the effect of supplementation on the course of the disease remains incompletely understood [18,19,20,21]. Discrepancies between the results of individual studies may result, among other factors, from differences in disease stage, disease-modifying treatment, baseline 25(OH)D concentration, the dose and supplementation regimen, and the endpoints used [19,20,21]. The results of D-Lay MS may suggest a potential role for intervention at a very early stage of the disease; however, differences in study populations and methodology do not allow direct comparison of its efficacy with supplementation in patients with established relapsing-remitting MS [19,20,21]. The available evidence does not currently support considering vitamin D an established, stand-alone disease-modifying treatment for MS [19,20,21].

4.2. Rheumatoid arthritis

Rheumatoid arthritis (RA) is a chronic systemic autoimmune disease in which persistent inflammation leads to progressive damage to joint structures. Vitamin D deficiency is frequently observed in patients with RA, but its significance in the pathogenesis of the disease remains unclear [9,22]. In a systematic review and meta-analysis, Clasen et al. found no convincing evidence that higher 25(OH)D concentrations were associated with a significantly lower risk of subsequently developing RA [9]. These findings therefore do not confirm vitamin D deficiency as an independent causal factor in the development of the disease.

More data are available on the potential association between vitamin D and the activity of established RA [9,22]. However, interventional studies of supplementation have yielded inconclusive results. The meta-analysis by Al-Saoodi et al. did not demonstrate consistent improvement across all assessed clinical and inflammatory parameters, although some analyses showed a beneficial effect on selected disease activity indices [22]. In contrast, the 2025 meta-analysis by Ranjbar et al. showed reductions in DAS28, CRP, and ESR after vitamin D supplementation, without significant improvement in function assessed using the HAQ or pain intensity measured on the VAS [23].

The available data therefore indicate that vitamin D supplementation may have a beneficial effect on some parameters of inflammatory activity in RA, but study results remain inconsistent [22,23]. The discrepancies may result, among other factors, from differences in baseline 25(OH)D concentration, supplementation dose and duration, and concomitant antirheumatic treatment [22,23]. The current evidence does not justify considering vitamin D supplementation a stand-alone disease-modifying therapy for RA [22,23].

4.3. Autoimmune thyroid diseases

The most common autoimmune thyroid diseases include Hashimoto's disease and Graves' disease. Patients with Hashimoto's disease frequently have lower vitamin D concentrations than healthy individuals, although the nature of this association has not been clearly established [24]. Given the immunomodulatory effects of vitamin D, its potential involvement in the regulation of the autoimmune response has been considered. However, the observed association does not allow the conclusion that vitamin D deficiency is an independent cause of disease development [24].

In Hashimoto's disease, data are also available on the effect of vitamin D supplementation on markers of autoimmunity. In the meta-analysis by Tang et al., which included 12 studies and 862 patients, vitamin D supplementation was associated with reduced titers of antibodies against thyroid peroxidase (anti-TPO) and thyroglobulin (anti-Tg) [25]. The authors also reported changes in selected parameters of thyroid function, with more favorable effects for some of the analyzed endpoints observed with a longer duration of supplementation [25]. A reduction in autoantibody titers is not, however, equivalent to a confirmed improvement in the long-term clinical course of the disease; therefore, the significance of these findings requires further investigation [24,25].

Less convincing data are available for Graves' disease. In the randomized, double-blind DAGMAR trial, which included 278 patients with newly diagnosed Graves' disease treated with antithyroid drugs, supplementation with 2800 IU of vitamin D daily did not increase the probability of achieving and maintaining remission compared with placebo [26]. The available data therefore indicate that low vitamin D concentrations are frequently observed in patients with Hashimoto's disease and that supplementation may affect selected markers of autoimmunity. However, a beneficial effect on the long-term clinical course of autoimmune thyroid diseases has not been conclusively demonstrated [24,25,26].

4.4. Inflammatory bowel disease

Inflammatory bowel disease (IBD), primarily comprising Crohn's disease (CD) and ulcerative colitis (UC), belongs to the group of chronic immune-mediated diseases of the gastrointestinal tract. Vitamin D deficiency is frequently observed in this group of patients; however, it remains unclear to what extent it is a consequence of the disease and to what extent it may participate in modulating its course [27]. Through the VDR receptor, vitamin D affects mucosal immune function, the integrity of the intestinal barrier, and the composition of the gut microbiota, providing a biological rationale for its potential importance in IBD [27]. Mechanistic data alone do not, however, allow confirmation of a causal relationship between vitamin D deficiency and the development of IBD [27].

The results of studies on supplementation are not fully consistent. In the meta-analysis by Valvano et al., which included 12 publications concerning randomized clinical trials, vitamin D supplementation was associated with a reduction in the overall risk of relapse or increased IBD activity (RR 0.64; 95% CI 0.46-0.89) [28]. The clearest effect was observed in patients with Crohn's disease who were in remission, in whom the risk of relapse was lower (RR 0.47; 95% CI 0.27-0.82) [28]. Data regarding UC were considerably more limited, and the effect of supplementation on continuous disease activity indices was inconclusive [28].

Caution in interpreting these findings is supported by the more recent meta-analysis by Ramezani et al., which included nine publications concerning randomized clinical trials. Supplementation effectively increased vitamin D concentrations; however, no significant effect on overall IBD activity or CRP concentration was demonstrated [29]. In contrast, in a large retrospective study published in 2026 that included 5,021 patients with IBD, vitamin D supplementation was associated with a lower frequency of disease-related hospitalizations and emergency department visits [30]. Because of the observational nature of this study, these findings do not allow confirmation of a causal relationship [30].

The available data therefore suggest that adequate vitamin D status may be relevant in patients with IBD and that supplementation may provide benefits in selected groups of patients, particularly those with Crohn's disease [28,29,30]. However, study results remain heterogeneous and do not currently provide sufficient grounds to consider vitamin D supplementation a stand-alone therapy that modifies the course of IBD [28,29,30].

5. Significance of vitamin D supplementation in autoimmune and immune-mediated diseases

In light of the available data, supplementation aimed at correcting vitamin D deficiency should be distinguished from attempts to use vitamin D as a specific immunomodulatory therapy [5,19,20,21,22,23,24,25,26,27,28,29,30]. Although experimental data indicate numerous mechanisms through which vitamin D affects the immune response, the optimal 25(OH)D concentration for achieving potential extraskeletal effects remains undetermined [5].

The results of interventional studies differ depending on the disease analyzed. In MS, some studies indicate a possible beneficial effect of supplementation on selected parameters of radiological activity, but an unequivocal reduction in relapse risk has not been demonstrated [19,20,21]. In RA, improvements in some indices of inflammatory activity have been observed, but the results of meta-analyses remain inconsistent [22,23]. In Hashimoto's disease, supplementation may reduce thyroid autoantibody titers, although the clinical significance of this effect remains uncertain, whereas in Graves' disease no increase in the probability of achieving and maintaining remission has been demonstrated [25,26]. In IBD, results regarding the effect of supplementation on disease course also remain heterogeneous [28,29,30].

When vitamin D deficiency is identified, its appropriate correction remains justified in accordance with general principles of supplementation; however, the available data do not support the routine use of high-dose vitamin D as a stand-alone treatment for autoimmune or immune-mediated diseases [5,19,20,21,22,23,25,26,28,29,30]. The supplementation regimen should take into account the individual clinical situation of the patient, including the severity of deficiency, age, body weight, comorbidities, and concomitant treatment [5].

6. Discussion

The available data indicate that vitamin D may participate in the regulation of numerous mechanisms relevant to the development of the autoimmune response. It affects dendritic cells, the balance between Th1 and Th17 lymphocytes and Treg cells, as well as B-lymphocyte function, providing a biological basis for a possible association between vitamin D deficiency and autoimmunity [7,15,16,17]. In inflammatory bowel disease, an additional role may be played by its effect on intestinal barrier integrity and the microbiota [27]. Mechanistic data do not, however, establish that vitamin D deficiency is a causal factor in the diseases analyzed.

In multiple sclerosis, an association between vitamin D status and disease activity has repeatedly been observed, but the results of supplementation studies remain inconclusive [18,19,20,21]. SOLAR and VIDAMS did not meet their primary endpoints, whereas in the D-

Lay MS study high-dose supplementation reduced the risk of composite disease activity in patients at a very early stage of the disease, with significant differences observed primarily in MRI parameters [19,20,21]. These findings raise the hypothesis that the significance of supplementation may depend, among other factors, on the timing of intervention; however, methodological differences between the studies do not allow unequivocal conclusions [19,20,21].

A similar heterogeneity applies to RA. It has not been convincingly confirmed that higher 25(OH)D concentrations are associated with a lower risk of developing the disease [9], whereas some interventional studies suggest a beneficial effect of supplementation on DAS28 and selected laboratory markers of inflammation [22,23]. In Hashimoto's disease, the observed reduction in anti-TPO and anti-Tg titers after supplementation may indicate an effect of vitamin D on selected markers of autoimmunity, but the significance of this effect for the long-term clinical course remains unclear [24,25]. In Graves' disease, the DAGMAR trial did not demonstrate an increased probability of achieving and maintaining remission after vitamin D supplementation [26]. This highlights the uncertainty regarding the clinical significance of supplementation in autoimmune thyroid diseases [24,25,26].

The results of studies in IBD are also not fully consistent. Some data indicate a possible reduction in relapse risk, particularly in patients with Crohn's disease who are in remission, whereas other analyses did not demonstrate a significant effect of supplementation on overall disease activity or CRP [28,29]. Beneficial associations observed in large retrospective studies require cautious interpretation because a causal relationship cannot be unequivocally established [30].

An important issue therefore remains the direction of the observed associations. Low 25(OH)D concentrations may potentially affect the immune response, but they may also be a consequence of chronic disease, lifestyle changes, reduced exposure to sunlight, or malabsorption, particularly in gastrointestinal diseases [7,27]. Heterogeneity of the results may additionally be related to differences in baseline vitamin D concentration, supplementation dose and duration, disease activity, concomitant treatment, and the endpoints used [5,19,20,21,22,23,28,29].

In summary, vitamin D deficiency is frequently observed in the autoimmune and immune-mediated diseases analyzed [7,22,24,27], and the biological basis for its potential significance is well justified. However, clinical data do not allow vitamin D deficiency to be unequivocally regarded as a cause of these diseases, nor supplementation as a specific immunomodulatory treatment. Correction of an identified deficiency remains the most justified approach, whereas determining whether supplementation can modify the natural course of specific diseases requires further well-designed interventional studies [5,19,20,21,22,23,25,26,28,29,30].

7. Conclusions

1. Vitamin D exerts immunomodulatory effects on both innate and adaptive immunity, providing a biological rationale for its potential significance in autoimmune and immune-mediated diseases.
2. Vitamin D deficiency is frequently observed in patients with multiple sclerosis, rheumatoid arthritis, autoimmune thyroid diseases, and inflammatory bowel diseases; however, the available data do not allow its causal role to be clearly established.
3. The results of studies on vitamin D supplementation remain heterogeneous. In some diseases, beneficial effects on selected laboratory parameters or disease activity indices have

been observed; however, it has not been conclusively demonstrated that supplementation modifies their long-term clinical course.

4. Correction of confirmed vitamin D deficiency remains justified, whereas current evidence does not support considering vitamin D supplementation as a stand-alone immunomodulatory therapy.

5. Further well-designed clinical studies are needed to determine whether vitamin D supplementation may have therapeutic significance in selected groups of patients.

Disclosure

Author Contributions

Conceptualization: Magdalena Iwaszko , Marcelina Skajewska,
Aleksandra Wojtera

Methodology: Szymon Bojanowski, Wojciech Ziębka

Literature search and literature review: Magdalena Iwaszko , Marcelina Skajewska

Writing rough preparation: Aleksandra Wojtera , Magdalena Iwaszko

Writing review and editing: Magdalena Iwaszko , Marcelina Skajewska,
Aleksandra Wojtera , Szymon Bojanowski, Wojciech Ziębka

Visualization: Szymon Bojanowski, Wojciech Ziębka

Supervision: Magdalena Iwaszko , Marcelina Skajewska,
Aleksandra Wojtera

Project administration: Magdalena Iwaszko

All authors have read and agreed to the published version of the manuscript.

Receiving funding: Not applicable - no specific funding.

Funding Statement:

The study did not receive external funding.

Institutional Review Board Statement:

Not applicable.

Informed Consent Statement:

Not applicable.

Data Availability Statement:

Not applicable.

Acknowledgments:

Not applicable.

Conflicts of Interest:

The authors declare no conflicts of interest.

Declaration of the Use of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used ChatGPT for the purpose of language editing and stylistic refinement to improve clarity and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the accuracy, integrity and conclusions of the publication.

References:

1. Martens, P. J., Gysemans, C., Verstuyf, A., & Mathieu, C. (2020). Vitamin D's effect on immune function. *Nutrients*, 12(5), 1248. <https://doi.org/10.3390/nu12051248>
2. Gallo, D., Baci, D., Kustrimovic, N., et al. (2023). How does vitamin D affect immune cells crosstalk in autoimmune diseases? *International Journal of Molecular Sciences*, 24(5), 4689. <https://doi.org/10.3390/ijms24054689>
3. Meza-Meza, M. R., Ruiz-Ballesteros, A. I., & de la Cruz-Mosso, U. (2022). Functional effects of vitamin D: From nutrient to immunomodulator. *Critical Reviews in Food Science and Nutrition*, 62(11), 3042–3062. <https://doi.org/10.1080/10408398.2020.1862753>
4. Brown, G., Marchwicka, A., & Marcinkowska, E. (2024). Vitamin D and immune system. *Advances in Food and Nutrition Research*, 109, 1–41. <https://doi.org/10.1016/bs.afnr.2023.12.00>
5. Giustina, A., Bilezikian, J. P., Adler, R. A., Banfi, G., Bikle, D. D., Binkley, N., Bollerslev, J., Bouillon, R., Brandi, M. L., Casanueva, F. F., di Somma, C., Erfurth, E. M., Fuleihan, G. E.-H., Fassio, A., Giustina, G., Marcocci, C., Messina, C., Napoli, N., Procopio, M., ... Cianferotti, L. (2024). Consensus statement on vitamin D status assessment and supplementation: Whys, whens, and hows. *Endocrine Reviews*, 45(5), 625–654. <https://doi.org/10.1210/endrev/bnae009>
6. Cui, A., Zhang, T., Xiao, P., Fan, Z., Wang, H., & Zhuang, Y. (2023). Global and regional prevalence of vitamin D deficiency in population-based studies from 2000 to 2022: A pooled analysis of 7.9 million participants. *Frontiers in Nutrition*, 10, 1070808. <https://doi.org/10.3389/fnut.2023.1070808>

7. Fletcher, J., Bishop, E. L., Harrison, S. R., Swift, A., Cooper, S. C., Dimeloe, S. K., Raza, K., & Hewison, M. (2022). Autoimmune disease and interconnections with vitamin D. *Endocrine connections*, 11(3), e210554. <https://doi.org/10.1530/EC-21-0554>
8. Martínez-Lapiscina, E. H., Mahatanan, R., Lee, C. H., Charoenpong, P., & Hong, J. P. (2020). Associations of serum 25(OH) vitamin D levels with clinical and radiological outcomes in multiple sclerosis: A systematic review and meta-analysis. *Journal of the Neurological Sciences*, 411, 116668. <https://doi.org/10.1016/j.jns.2020.116668>
9. Clasen, J. L., Cole, R., Aune, D., Sellon, E., & Heath, A. K. (2023). Vitamin D status and risk of rheumatoid arthritis: systematic review and meta-analysis. *BMC rheumatology*, 7(1), 3. <https://doi.org/10.1186/s41927-023-00325-y>
10. Islam, M. A., Khandker, S. S., Alam, S. S., Kotyla, P., & Hassan, R. (2019). Vitamin D status in patients with systemic lupus erythematosus (SLE): A systematic review and meta-analysis. *Autoimmunity Reviews*, 18(11), 102392. <https://doi.org/10.1016/j.autrev.2019.102392>
11. Gubatan, J., Chou, N. D., Nielsen, O. H., & Moss, A. C. (2019). Systematic review with metaanalysis: Association of vitamin D status with clinical outcomes in adult patients with inflammatory bowel disease. *Alimentary Pharmacology & Therapeutics*, 50(11–12), 1146–1158. <https://doi.org/10.1111/apt.15506>
12. Hahn, J., Cook, N. R., Alexander, E. K., et al. (2022). Vitamin D and marine omega 3 fatty acid supplementation and incident autoimmune disease: VITAL randomized controlled trial. *BMJ*, 376, e066452. <https://doi.org/10.1136/bmj-2021-066452>
13. Nguyen, M., Bryant, K., & O'Neill, S. (2018). Vitamin D in SLE: A role in pathogenesis and fatigue? A review of the literature. *Lupus*, 27(13), 2003–2011. <https://doi.org/10.1177/0961203318796293>
14. Carlberg, C. (2019). Vitamin D signaling in the context of innate immunity: Focus on human monocytes. *Frontiers in Immunology*, 10, 2211. <https://doi.org/10.3389/fimmu.2019.02211>
15. Hafkamp, F. M. J., Taanman-Kueter, E. W. M., van Capel, T. M. M., Kormelink, T. G., & de Jong, E. C. (2022). Vitamin D3 Priming of Dendritic Cells Shifts Human Neutrophil-Dependent Th17 Cell Development to Regulatory T Cells. *Frontiers in immunology*, 13, 872665. <https://doi.org/10.3389/fimmu.2022.872665>
16. Nagy, N. A., Lozano Vigario, F., Sparrius, R., van Capel, T. M. M., van Ree, R., Tas, S. W., de Vries, I. J. M., Geijtenbeek, T. B. H., Slütter, B., de Jong, E. C., & DC4Balance consortium (2023). Liposomes loaded with vitamin D3 induce regulatory circuits in human dendritic cells. *Frontiers in immunology*, 14, 1137538. <https://doi.org/10.3389/fimmu.2023.113753>
17. Ghaseminejad-Raeini, A., Ghaderi, A., Sharafi, A., Nematollahi-Sani, B., Moossavi, M., Derakhshani, A., & Sarab, G. A. (2023). Immunomodulatory actions of vitamin D in various immune-related disorders: a comprehensive review. *Frontiers in immunology*, 14, 950465. <https://doi.org/10.3389/fimmu.2023.950465>
18. Giordano, A., Clarelli, F., Pignolet, B., Mascia, E., Sorosina, M., Misra, K., Ferrè, L., Bucciarelli, F., Manouchehrinia, A., Moiola, L., Martinelli, V., Rocca, M. A., Liblau, R., Filippi, M., & Esposito, F. (2025). Vitamin D affects the risk of disease activity in multiple sclerosis. *Journal of neurology, neurosurgery, and psychiatry*, 96(2), 170–176. <https://doi.org/10.1136/jnnp-2024-334062>
19. Thouvenot, E., Laplaud, D., Lebrun-Frenay, C., Derache, N., Le Page, E., Maillart, E., Froment-Tilkete, C., Castelnovo, G., Casez, O., Coustans, M., Guennoc, A. M., Heinzlef, O.,

- Magy, L., Nifle, C., Ayrignac, X., Fromont, A., Gaillard, N., Caucheteux, N., Patry, I., De Sèze, J., ... D-Lay MS Investigators (2025). High-Dose Vitamin D in Clinically Isolated Syndrome Typical of Multiple Sclerosis: The D-Lay MS Randomized Clinical Trial. *JAMA*, 333(16), 1413–1422. <https://doi.org/10.1001/jama.2025.1604>
20. Hupperts, R., Smolders, J., Vieth, R., Holmøy, T., Marhardt, K., Schlupe, M., Killestein, J., Barkhof, F., Beelke, M., Grimaldi, L. M. E., & SOLAR Study Group (2019). Randomized trial of daily high-dose vitamin D3 in patients with RRMS receiving subcutaneous interferon β -1a. *Neurology*, 93(20), e1906–e1916. <https://doi.org/10.1212/WNL.00000000000008445>
21. Cassard, S. D., Fitzgerald, K. C., Qian, P., Emrich, S. A., Azevedo, C. J., Goodman, A. D., Sugar, E. A., Pelletier, D., Waubant, E., & Mowry, E. M. (2023). High-dose vitamin D3 supplementation in relapsing-remitting multiple sclerosis: a randomised clinical trial. *EClinicalMedicine*, 59, 101957. <https://doi.org/10.1016/j.eclinm.2023.101957>
22. Al-Saoodi, H., Kolahdooz, F., Andersen, J. R., & Jalili, M. (2024). Effect of vitamin D on inflammatory and clinical outcomes in patients with rheumatoid arthritis: a systematic review and dose-response meta-analysis of randomized controlled trials. *Nutrition reviews*, 82(5), 600–611. <https://doi.org/10.1093/nutrit/nuad083>
23. Ranjbar, M., Rahimlou, M., Fallah, M., Djafarian, K., & Mohammadi, H. (2025). Effects of vitamin D supplementation in patients with rheumatoid arthritis: A systematic review and meta-analysis. *Heliyon*, 11(3), e42463. <https://doi.org/10.1016/j.heliyon.2025.e42463>
24. Durá-Travé, T., & Gallinas-Victoriano, F. (2024). Autoimmune Thyroiditis and Vitamin D. *International journal of molecular sciences*, 25(6), 3154. <https://doi.org/10.3390/ijms25063154>
25. Tang, J., Shan, S., Li, F., & Yun, P. (2023). Effects of vitamin D supplementation on autoantibodies and thyroid function in patients with Hashimoto's thyroiditis: A systematic review and meta-analysis. *Medicine*, 102(52), e36759. <https://doi.org/10.1097/MD.00000000000036759>
26. Grove-Laugesen, D., Ebbeløj, E., Watt, T., Riis, A. L., Østergård, T., Bruun, B. J., Juel Christiansen, J., Hansen, K. W., & Rejnmark, L. (2023). Effect of Vitamin D Supplementation on Graves' Disease: The DAGMAR Trial. *Thyroid : official journal of the American Thyroid Association*, 33(9), 1110–1118. <https://doi.org/10.1089/thy.2023.0111>
27. Vernia, F., Valvano, M., Longo, S., Cesaro, N., Viscido, A., & Latella, G. (2022). Vitamin D in Inflammatory Bowel Diseases. Mechanisms of Action and Therapeutic Implications. *Nutrients*, 14(2), 269. <https://doi.org/10.3390/nu14020269>
28. Valvano, M., Magistroni, M., Cesaro, N., Carlino, G., Monaco, S., Fabiani, S., Vinci, A., Vernia, F., Viscido, A., & Latella, G. (2024). Effectiveness of Vitamin D Supplementation on Disease Course in Inflammatory Bowel Disease Patients: Systematic Review With Meta-Analysis. *Inflammatory bowel diseases*, 30(2), 281–291. <https://doi.org/10.1093/ibd/izac253>
29. Ramezani, R., Ghorbaninejad, P., Eslahi, M., Sheikhi, L., Abbasi, F., Hasanzadeh, M., Mohammadpour, S., & Milajerdi, A. (2024). Effects of Vitamin D Supplementation on Serum 25-Hydroxy Cholecalciferol in Inflammatory Bowel Diseases: A Meta-Analysis of Randomized Clinical Trials. *International journal of preventive medicine*, 15, 65. https://doi.org/10.4103/ijpvm.ijpvm_133_23
30. Sninsky, J. A., Sansgiry, S., Taylor, T., Perrin, M., Kanwal, F., & Hou, J. K. (2026). The Real-World Impact of Vitamin D Supplementation on Inflammatory Bowel Disease Clinical Outcomes. *Clinical gastroenterology and hepatology : the official clinical practice journal of the American Gastroenterological Association*, 24(6), 1666–1674.e2. <https://doi.org/10.1016/j.cgh.2025.07.013>